

National University of Computer and Emerging Sciences

Fast School of Computing

Spring 2026

CS-4031 Compiler Construction

Quiz # 4 [10 marks]

Section: _____

Time = 20 minutes

06/04/2026

Name: _____

Roll No: _____

Question 1: Give Short Answer for the following parts.

- 1) Which of the following grammars can be parsed by LR(0) but not by SLR(1), Give one line reason also?

A) $S \rightarrow a A, A \rightarrow b$

B) $S \rightarrow A a \mid b, A \rightarrow c$

C) $S \rightarrow a A a, A \rightarrow b$

Solution _____ None of them. If a grammar is LR(0) then it is also SLR(1)

- 2) Which of the following grammars will cause a shift-reduce conflict in an LR(0) parser, Give reason ?

A) $S \rightarrow a A, A \rightarrow b$

B) $S \rightarrow A a \mid b, A \rightarrow c$

C) $S \rightarrow a A b, A \rightarrow c$

D) $S \rightarrow A, A \rightarrow a, A \mid \epsilon$

Solution _____ D option. It contains $A \rightarrow \cdot$ which is a final item in state 0. It makes a S/R conflict.

- 3) For the grammar: $S \rightarrow A a \mid B, A \rightarrow b B, B \rightarrow c \mid \epsilon$. What is the DFA state that will exhibit a shift-reduce conflict?

Solution _____ For LR(0) grammar. State 0 exhibit a shift reduce conflict for items.

- 4) Consider the grammar:

$S \rightarrow A B, A \rightarrow a \mid \epsilon, B \rightarrow b \mid \epsilon$. With the augmented rule $S' \rightarrow S$, how many states are in the LR (0) DFA, accounting for all closures and transitions?

Solution _____ The DFA has 6 states

- 5) Consider the Grammar

$S \rightarrow A a \mid B a$

$A \rightarrow c A \mid d$

$B \rightarrow c B \mid d$

In SLR(1), a reduce-reduce conflict occurs in a state with $[A \rightarrow d \cdot]$ and $[B \rightarrow d \cdot]$. Why does this conflict persist despite follow sets?

Solution _____ Because Follow sets of A and B both contains symbol a.

- 6) Consider the following grammar: Is it LR(0) or SLR(1) or Both give reason?
 $S \rightarrow aAc \mid aBd$
 $A \rightarrow b$
 $B \rightarrow b$

Solution _____ The grammar is not LR(0) as a state will contain $A \rightarrow b$. and $B \rightarrow b$. both exhibiting a conflict. It is SLR(1) as Follow (A) = {c} and Follow(B) = {d}, hence no conflict

Question 3: Consider the Following Grammar.

Consider the following Grammar

$S' \rightarrow S$
 $S \rightarrow A b \mid a A$
 $A \rightarrow a A \mid \epsilon$

Construct the DFA of LR(0) items and Fill the SLR (1) table for the above grammar?

State	a	b	\$	Goto S	Goto A
I0					
I1					
I2					
I3					
I4					
I5					
I6					